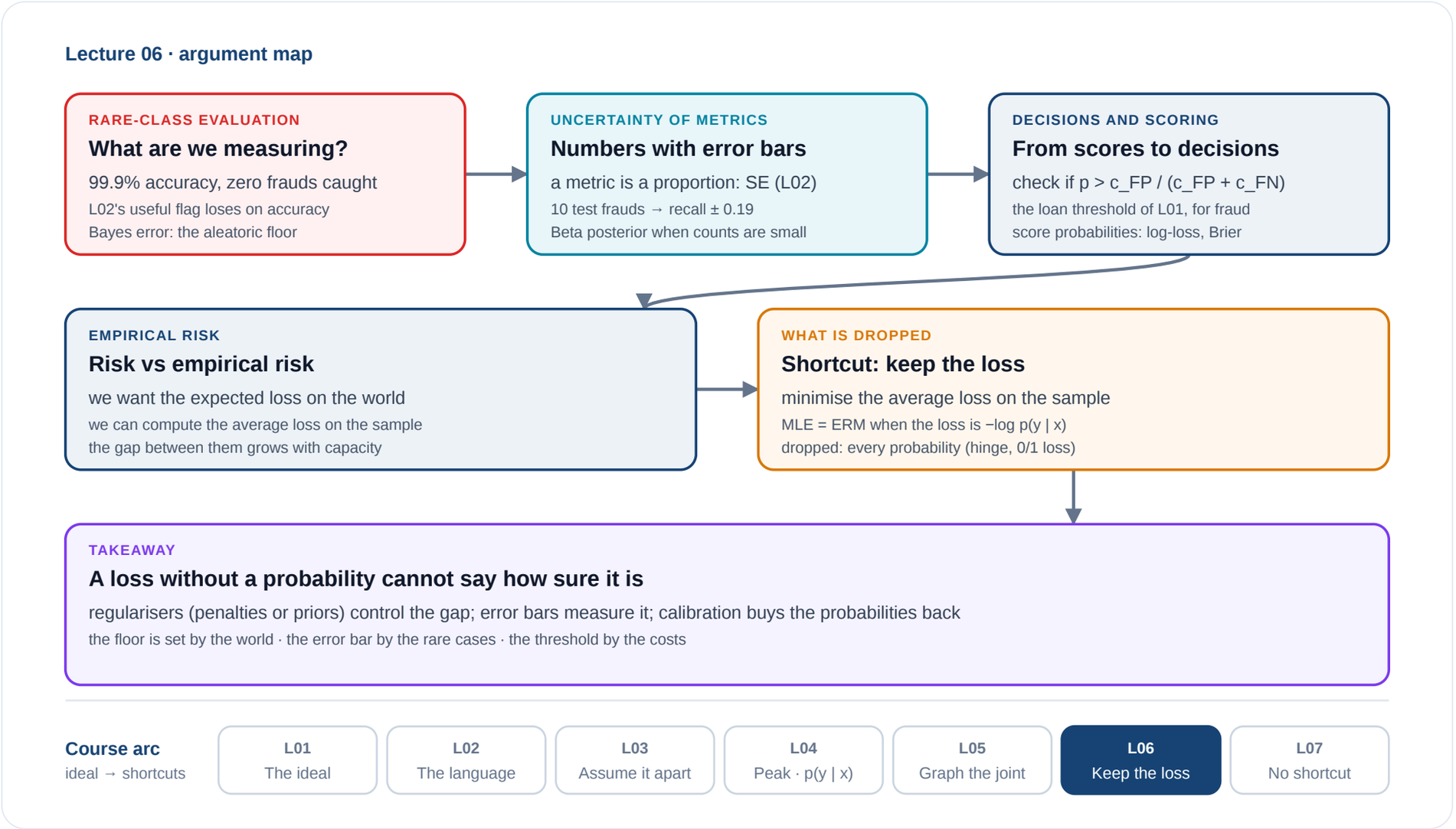
Probabilistic Machine Learning

Lecture 06: Evaluation and Empirical Risk Minimisation

Prof. Dr. Michael Granitzer



Argument Map



Evaluating Rare-Class Classifiers

Guiding Question: A fraud model reports 99.9% accuracy and has never caught a single fraud. What should we have measured instead — and how low could the error of the best model be?

What We Want to Know

The **true error** of a classifier h is the probability that it errs on a fresh example drawn from the world:

$$\mathrm{err}^*(h) = P\big(h(X) \neq Y\big), \qquad (X, Y) \sim p$$

The **Bayes error** is the smallest true error any classifier can reach:

$$\min_h \mathrm{err}^*(h) = \mathbb{E}_X\Big[\min\big(p(y{=}1 \mid X),\ 1 - p(y{=}1 \mid X)\big)\Big]$$

A transaction type is fraudulent 55% of the time. Could a better model push its error below 45%?

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No — only **new attributes** can. The Bayes error is aleatoric uncertainty (Lecture 01's identical patients) written as an error rate: the floor that no amount of data removes.

Lecture 02's Flag, Evaluated

per 100,000 transactions, 100 frauds	always "legitimate"	Lecture 02's flag
frauds caught · honest customers flagged	0 · 0	99 · 999
accuracy	**99.9%**	**99.0%**
recall $P(\text{flag} \mid \text{fraud})$	0%	99%
precision $P(\text{fraud} \mid \text{flag})$	—	9%

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Which model would you deploy — and which one does accuracy recommend?

> By accuracy, the model that catches **nothing** wins. On a rare class, accuracy counts the majority and hides that a model is useless.

What to Report Instead

- → **A baseline** next to every number: always-majority for classification, the mean for regression. "99.9%" is not a result until you know that doing nothing scores 99.9%.
- → **Measures that condition on the rare class**: recall conditions on the truth, precision on the decision — the two directions of Bayes' theorem that Lecture 02 warned not to swap.
- → **The costs**: which of the two mistakes the application pays for. The decisions module turns them into a threshold.

> Precision depends on the base rate, recall does not: the same flag has precision 9% at 1 in 1,000 and about 50% at 1 in 100.

Mini-Exercise

A model flags **every** transaction as fraud (one in 1,000 is fraudulent).

What are its recall and its precision?

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Recall **100%** — it catches every fraud. Precision **0.1%** — the base rate. (Accuracy: 0.1%.)
Recall alone can be bought by flagging everything; that is why it is always reported together with precision, or with the costs.

Takeaway

We want the **true error**, whose floor — the **Bayes error** — is aleatoric uncertainty that no data removes. On a rare class, accuracy measures the majority: report a **baseline** and measures that condition on the rare class.

Uncertainty of Metrics

Guiding Question: The test set says recall 0.90. Would you promise the bank that the model catches 90% of next month's frauds?

Where an Error Number Comes From

estimate	how	what it is
training error	evaluate on the data the model was fitted on	optimistically biased — on average below the true error of the fitted model, often far below
hold-out	split once; fit on one part, evaluate on the other	an unbiased estimate of err^* — for **one** split, with a sampling error
k-fold cross-validation	each of k folds serves once as the test set; average	uses every example for testing once; still an estimate from a sample

> A test set that is consulted for choices — which model, which threshold — becomes training data. Keep one set that is looked at once.

A Metric Is a Proportion

Accuracy on n test cases is the mean of n yes/no indicators — a Bernoulli sample (Lecture 02). Its standard error is $\sigma/\sqrt{n}$ with $\sigma^2 = p(1-p)$:

$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

A test set of **10,000** transactions, one in 1,000 fraudulent:

0.990 ± 0.002

accuracy: $n = 10{,}000$, $\mathrm{SE} \approx 0.001$

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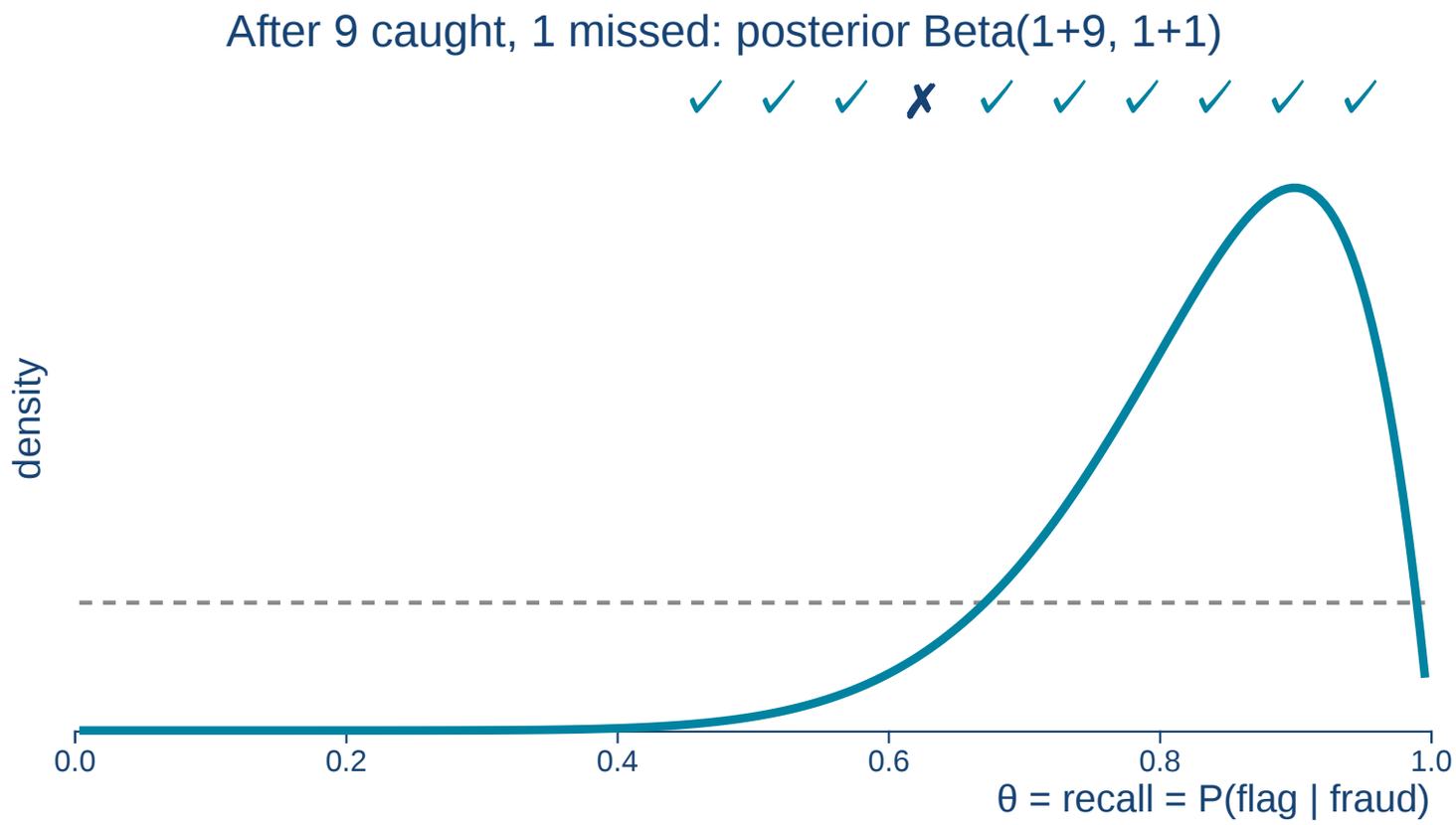
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Same test set, same model — why is one error bar a hundred times wider?

Recall is a proportion **of the frauds**, and there are only ten of them. The error bar is set by the rare class, not by the size of the test set.

Recall as an Unknown Parameter



After 9 caught and 1 missed, with a flat prior: $\mathrm{Beta}(10, 2)$ — mean **0.83**, 95% credible interval **[0.59, 0.98]**.

The Rare Class Sets the Test Size

Recall near 0.9, so $\mathrm{SE} = \sqrt{0.09/n_{\mathrm{fraud}}}$; at one fraud in 1,000 transactions:

frauds in the test set	transactions	SE of recall	error bar (±2 SE)
10	10,000	0.095	±0.19
100	100,000	0.030	±0.06
400	400,000	0.015	±0.03
1,000	1,000,000	0.0095	±0.02

Model B catches 10 of the 10 test frauds, model A catches 9. That is **one fraud** of difference — well inside either error bar. Compare two models on the **same** test cases and look at the cases where they disagree.

Mini-Exercise

Recall is near 0.9, so $\mathrm{SE} = \sqrt{0.09/n}$ for n test frauds.

How many frauds does the test set need for $\mathrm{SE} = 0.03$ — and how many transactions is that?

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Recall is near 0.9, so $\mathrm{SE} = \sqrt{0.09/n}$ for n test frauds.

How many frauds does the test set need for $\mathrm{SE} = 0.03$ — and how many transactions is that?

$0.09/n = 0.0009$, so $n = 100$ frauds — about **100,000 transactions** at one in 1,000. For $\mathrm{SE} = 0.015$ it would be four times as many.

Takeaway

> Every metric is an **estimate from a sample** and needs an **error bar**: a standard error, or a Beta posterior when the counts are small. For a rare class the error bar is set by the number of **rare cases**, not by the size of the test set.

Decisions and Scoring Rules

Guiding Question: The model says p(fraud) = 0.09. Check the transaction or let it through — and who decides where the line is?

Cost Threshold (Recap)

	transaction is fraud	transaction is legitimate
check it (a text to the customer)	0	€5 (c_{FP})
let it through	€500 (c_{FN})	0

From Lecture 03: check iff

$$p(\text{fraud} \mid \mathbf{x}) > p^* = \frac{c_{FP}}{c_{FP} + c_{FN}} = \frac{5}{505} \approx 0.01$$

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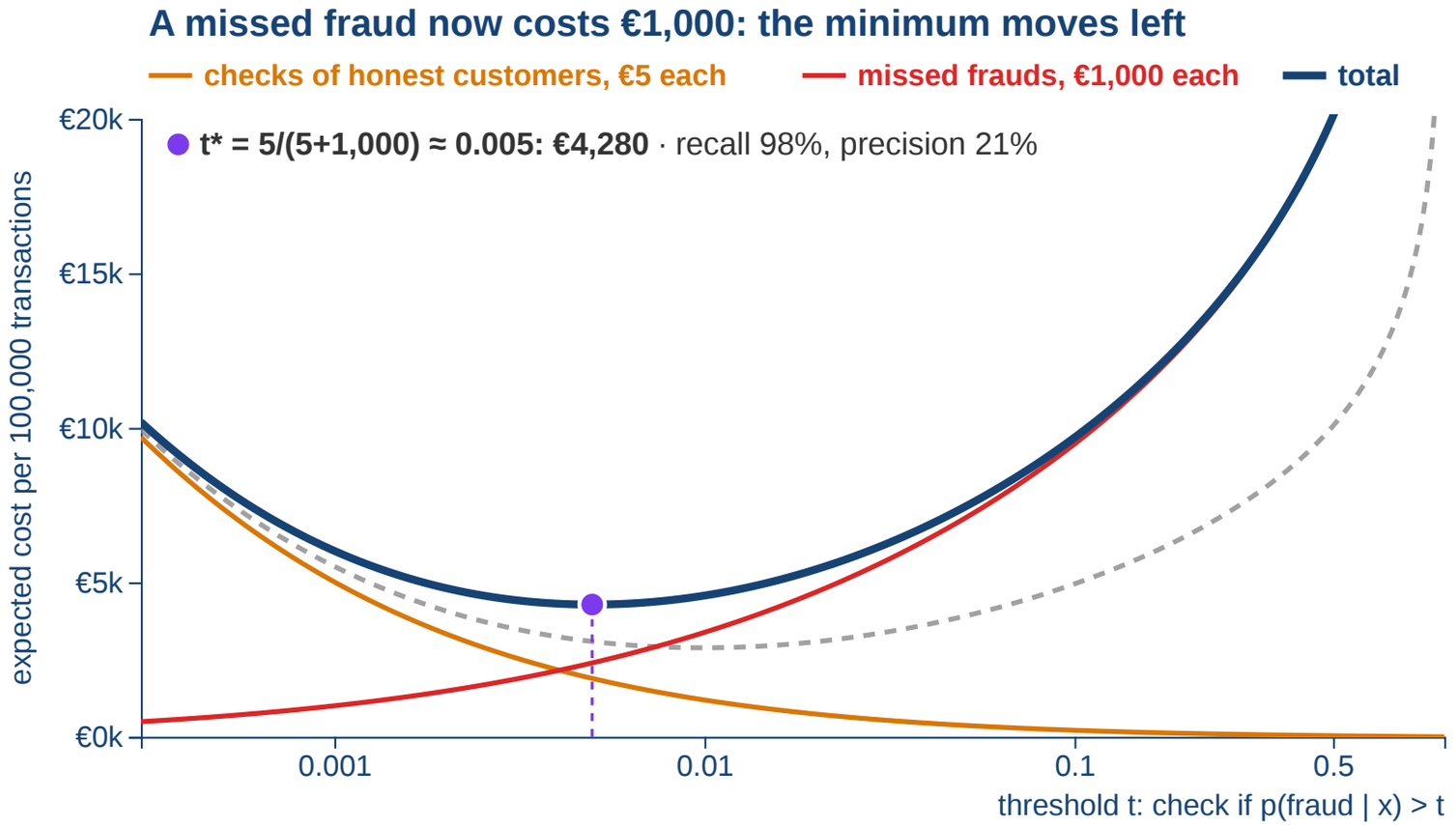
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> Yes: 9% is nine times the break-even. The criterion is expected cost, not precision.

Sweeping the Threshold



Predict before the last click: a missed fraud now costs €1,000. Does the best threshold move left or right — and does recall go up or down?

Scoring the Probabilities Themselves

A threshold uses the score once. To evaluate the **probabilities**, score every prediction on held-out data:

Log-loss

$$\text{log-loss} = -\frac{1}{n}\sum_{i=1}^{n} \log p_{\text{model}}(y_i \mid \mathbf{x}_i)$$

the held-out negative log-likelihood — Lecture 04's cross-entropy, on test data

Brier score

$$\text{Brier} = \frac{1}{n}\sum_{i=1}^{n} \left(p_i - y_i\right)^2$$

squared error of the probability against the 0/1 outcome

Both scores are **proper**: in expectation they are best when the model reports the true probability. Accuracy is not strictly proper — it cannot tell 0.51 from 0.99.

What the Scores Punish

the model said $p = 0.09$, and the transaction was…	log-loss term	Brier term
fraud	2.41	0.83
legitimate	0.09	0.008
fraud, but the model had said $p = 0.001$	**6.91**	0.998

Log-loss punishes **confident errors** without limit — $p = 0$ on a fraud costs infinity; Brier is bounded by 1. Whether "0.09" comes true 9% of the time is Lecture 04's **reliability diagram**, now drawn on held-out data.

Mini-Exercise

Checking costs €5, a missed fraud €500. The model says $p = 0.02$.

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Checking costs €5, a missed fraud €500. The model says $p = 0.02$.

Check the transaction or let it through?

Check. Letting it through costs 0.02 · €500 = €10 in expectation; checking costs 0.98 · €5 ≈ €4.90. Equivalently, $0.02 > p^* \approx 0.01$.

Takeaway

A **calibrated probability** plus the **costs** gives the decision: act when $p > c_{FP}/(c_{FP} + c_{FN})$. Precision and recall move along one curve as the threshold moves; **log-loss** and the **Brier score** evaluate the probabilities themselves.

Empirical Risk Minimisation

Guiding Question: Every model you trained minimised something on the training data. What did we actually want — and what did the substitution throw away?

What We Actually Want to Minimise

The **risk** of a hypothesis h under a loss L is its expected loss on the world:

$$R(h) = \mathbb{E}_{(\mathbf{x},y)\sim p}\big[L\big(h(\mathbf{x}), y\big)\big]$$

0/1 loss

$R(h) = \mathrm{err}^*(h)$ — the true error of the rare-class module

Cost matrix

$R(h)$ = the expected cost of the decisions module

It is the quantity that matters — and **uncomputable**: p is unknown.

What Training Computes Instead

The **empirical risk** is the average loss on the training sample, and **empirical risk minimisation (ERM)** picks its minimiser:

$$\hat{R}(h) = \frac{1}{n}\sum_{i=1}^{n} L\big(h(\mathbf{x}_i), y_i\big), \qquad \hat{h} = \arg\min_{h \in \mathcal{H}} \hat{R}(h)$$

loss	trained model	is $h(\mathbf{x})$ a probability?
squared error	linear regression	the mean of a Gaussian — **yes** (Lecture 04)
cross-entropy $-\log p(y \mid \mathbf{x})$	logistic regression, most neural nets	**yes** — this is maximum likelihood for $p(y \mid \mathbf{x})$
hinge $\max(0, 1 - y \cdot h(\mathbf{x}))$	support vector machine	**no** — a signed margin score
0/1	what we care about; no useful gradient	**no** — only a label

Keep the Loss, Drop the Probabilities

WHAT ERM WITH A HINGE OR 0/1 LOSS RETURNS

a score: 2.3

large means "more fraud-like" — nothing more

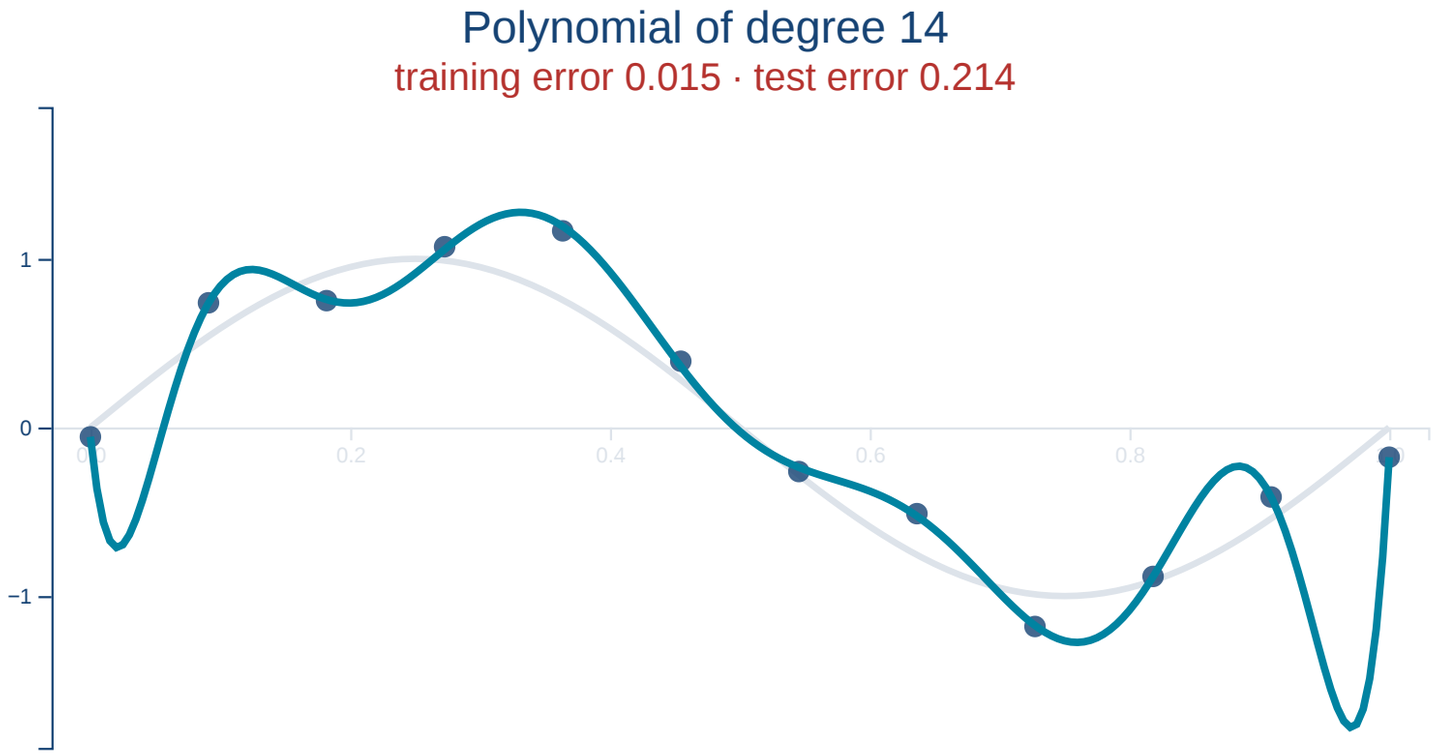
WHAT IT CAN NO LONGER DO

- apply the cost threshold $p > c_{FP}/(c_{FP} + c_{FN})$ of the decisions module
- be scored by log-loss or Brier, or drawn in a reliability diagram
- put an error bar on its own confidence

To use it for a decision, **buy the probabilities back**: fit $p = \sigma(a \cdot \text{score} + b)$ on held-out data (Platt scaling) — a logistic regression on the score, i.e. calibration.

Not every method is ERM: maximum likelihood for $p(y \mid \mathbf{x})$ is ERM with loss $-\log p$; generative maximum likelihood (naive Bayes) scores $\log p(\mathbf{x}, y)$, and Bayesian methods minimise nothing.

The Generalisation Gap



$R(\hat{h}) = \hat{R}(\hat{h}) + \big(R(\hat{h}) - \hat{R}(\hat{h})\big)$: the gap opens because $\hat{h}$ was **chosen on the data that then measures it**, and it tends to grow with the capacity of $\mathcal{H}$.

Structural Risk Minimisation — and the Prior Inside It

Minimise the empirical risk **plus** a complexity penalty:

$$\hat{h} = \arg\min_{h \in \mathcal{H}} \hat{R}(h) + \lambda \Omega(h)$$

With the loss $-\log p(y \mid \mathbf{x}, \mathbf{w})$, the MAP estimate of Lecture 04 has exactly this form:

$$\frac{1}{n} \sum_{i=1}^{n} -\log p(y_i \mid \mathbf{x}_i, \mathbf{w}) \quad \underbrace{-\frac{1}{n} \log p(\mathbf{w})}_{\lambda \Omega(\mathbf{w})}$$

> The $\frac{1}{n}$ is the point: the prior's weight **fades as data grows** — the pseudo-counts of Lecture 04's Beta prior. A Gaussian prior $\mathcal{N}(0, \tau^2 \mathbf{I})$ gives $\lambda \Omega(\mathbf{w}) = \|\mathbf{w}\|^2 / (2n\tau^2)$.

Mini-Exercise

A linear SVM gives a transaction the score **2.3**. The fraud team checks every transaction with $p > 0.01$.

Can they apply their rule to this transaction?

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Can they apply their rule to this transaction?

Not directly. 2.3 is a margin score, not a probability — the hinge loss dropped it. First calibrate on held-out data, $p = \sigma(a \cdot 2.3 + b)$, then compare that p with 0.01.

Takeaway

Training minimises the **empirical risk** because the **risk** is out of reach. ERM keeps the **loss** and — unless the loss is a negative log-probability — drops the **probabilities**. Regularisers (penalties or priors) control the gap; error bars measure it; calibration buys the probabilities back.